MUTHUKUMARAN YOGEESWARAN

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EXPERIENCE

Robotics Perception Intern, DSO National Laboratories **Sep 2026 – Nov 2026**

- Built a **real-time**, camera-only perception pipeline for **maritime robots**, covering **obstacle detection, tracking and depth estimation**.
- Adapted state-of-the-art **open-vocabulary vision models** to maritime scenes, so obstacles **not seen in training** are still detected.

Research Student, NUS Immersive Reality Lab **Aug 2025 – May 2026**

- Built a **fully offline companion robot** for eldercare with **NUS Nursing**, running the whole voice and perception stack (VAD, ASR, **on-device LLM**, TTS) on one **NVIDIA Jetson**.
- Added **YOLO-based face tracking** and **emotion recognition** so the robot reads the user's mood and reacts in real time, with digital facial expressions and head movement on a custom **3D-printed 2-DOF head**.

Robot Operator, Cortex AI **Nov 2025 – Jan 2026**

- Collected **Vision-Language-Action (VLA)** training data via teleoperated **YAM robot arms** and **egocentric data capture with a Meta Quest 3**, designing new tasks to expand dataset coverage.
- Worked hands-on with the **LeRobot** dataset framework and codebase, evaluating **data quality** and separating usable demonstrations from low-quality or noisy ones.

Research Intern, The University of Osaka **May 2025 – Aug 2025**

- Built a real-time **human localisation system** for a social robot: trained a **custom YOLO keypoint model** and fused its detections into the robot's **SLAM map** via Kabsch alignment, reaching **2–6 cm** localisation **accuracy**.
- **Deployed live at Osaka Expo 2025** (Yoshikawa Lab pavilion), localising visitors in crowds so the robot could meet their gaze.

Research Intern, A*STAR (Advanced Remanufacturing & Technology Centre) **May 2024 – Aug 2024**

- Trained a **dexterous robotic hand** to grasp and relocate objects in **MuJoCo physics simulation** using **DAPG**, a reinforcement learning method bootstrapped from human demonstrations.
- Adapted the entire **simulation pipeline** from a research hand (Adroit) to the 6-actuator Inspire hand, converting the robot model from URDF to MJCF and remapping the observation space.

PROJECTS

BoxBunny: AI Boxing Sparring & Coaching Robot (Final-Year Capstone) **Aug 2025 – May 2026**

- Led robot intelligence for an adaptive boxing robot, achieving **96.8% punch recognition at 30 fps** on unseen users with a custom **transformer model** optimised with **TensorRT** on a **Jetson Orin NX**.
- Architected the **ROS 2** runtime connecting perception, action recognition and an **LLM-based coaching loop**, hardened with unit and integration testing.
- Curated a custom **RGB-D action dataset** with an **Intel RealSense D435i** and built an in-house annotation GUI for labelling and evaluation.
- Presented the project at the **International Senior Project Conference 2026** (KMUTT, Bangkok) and demoed it live at the NUS CDE AI Fair.

Breaking Bias: Bias Evaluation & Mitigation in LLMs (University of Toronto exchange) **Jan 2025 – May 2025**

- Investigated how **biased and easily manipulated** LLMs are when used to **score essays**, attacking them with **prompt injection**, word-embedding noise and sentence scrambling.
- Tested whether a **committee of different LLMs** (Phi-4-mini, Llama 3.2) scores more fairly than one model.

NUS RoboMaster: Two-Legged Self-Balancing Robot (DJI RoboMaster Championships) **Aug 2023 – Dec 2024**

- Software team member in a team of 4: **simulated the robot dynamics and tuned LQR/PID control in Simulink** before deploying to **STM32**.
- Contributed to the chassis **CAD design** and prototyped a **computer vision aim-bot** for auto-targeting.

TECHNICAL SKILLS & CERTIFICATIONS

- **Programming Languages: Python**, **C++**, C, MATLAB, JavaScript, Assembly
- **Machine Learning & AI: PyTorch**, **Computer Vision**, Neural Networks, Action Recognition (Pose / Temporal Models), **Reinforcement Learning**, Large Language Models (LLMs) + RAG, TensorRT
- **Robotics & Systems: ROS 2**, SLAM, **Robot Perception**, Teleoperation (LeRobot), Mobile Robotics, **NVIDIA Jetson**, Intel RealSense, STM32
- **Simulation & Control: MuJoCo**, MATLAB / Simulink, **PID / LQR Control**, COMSOL Multiphysics, SolidWorks Simulation
- **CAD & Design:** SolidWorks (**CSWA**, Mechanical Design & Additive Manufacturing)
- **Tools & Platforms:** Git, GitHub, **Docker**, Conda, **Linux (Ubuntu)**

EDUCATION

National University of Singapore, Bachelor of Engineering **Aug 2022 – Jun 2026**

- **Honours (Distinction)**, GPA **4.35 / 5.00**
- Major in Engineering Science with a specialisation in **Computational Engineering Science**
  - Minors in Innovation & Design and Computing (Design and Engineering)
- Exchange semester at the **University of Toronto** (Jan 2025 – May 2025)
- Research internship at the **University of Osaka** (May 2025 – Aug 2025)
- Focus areas: **Robotics, Artificial Intelligence, Human–Robot Interaction**

LEADERSHIP & ACTIVITIES

President, NUS Engineering Science Student Wing **Aug 2023 – Aug 2024**

- Led a **40-member team** planning events for the Engineering Science student body.
- Previously on the planning committee for **MakerMania23**, a national prototyping competition for teams from Junior Colleges, Polytechnics and Universities.

NUS Student Ambassador, National University of Singapore **Jan 2023 – May 2026**

- Represented NUS in communications, outreach and official university events.